Statistical Efficiency of Birth-Death Tree Parameter Estimation

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Source code available at:
PhylodynamicsDL: https://github.com/mcanearm/PhylodynamicsDL
PhyloDeepPOMP: https://github.com/Horopter/PhyloDeepPOMP

1 Intro

[**voznica˙deep˙2022**]

The Phylodeep package implements a deep learning approach to estimating parameters of several different tree models. The results of the deep learning models are impressive, but noticeably absent from the baseline of the models is the closed-form MLE for any of the simulated trees. For our study, we focused on the Birth-Death trees and compared the statistical efficiency of the Phylodeep approach to the traditional Maximum Likelihood Estimation (MLE) approach. Initial comparisons indicated that there may be some issues with the likelihood estimation component, but we think the results are interesting enough to present, albeit with caveats.

2 Data Generation

The original paper by Voznicka et. al.one of the other authors. We chose to replicate this process as faithfully as possible, both for reasons of simplicity and reproducibility.

Listing 1: Simulator Code

```
subprocess.run(
    [
        str(generate_bd), "--min_tips", "10", "--max_tips", "500",
            "--la", str(la), "--psi", str(psi), "--p", str(0.5),
            "--nwk", tree_path, "--log", log_path,
    ],
    check=True,
    timeout=timeout_seconds,
    stdin=subprocess.DEVNULL,
    stdout=subprocess.DEVNULL,
    stderr=subprocess.DEVNULL,
)
```

This is identical to the process outlined in the Phylodeep with the exception that we reduce the minimum tips to 10. Our random sampling of λ and ψ are between 0 and 1. These ranges do not align with the Phlyodeep paper in the supplementary data, and we locked the sampling probability at 0.5 rather than sampling it as well. By focusing on two parameters, we are simplifying our analysis further, which seemed reasonable given the timeframe of the project. However,

3 Methods

4 MLE

We made a few attemps at implementing the MLE, which ended up being a more challenging part of the project than we expected, mostly due to difficulties in dealing with the data.

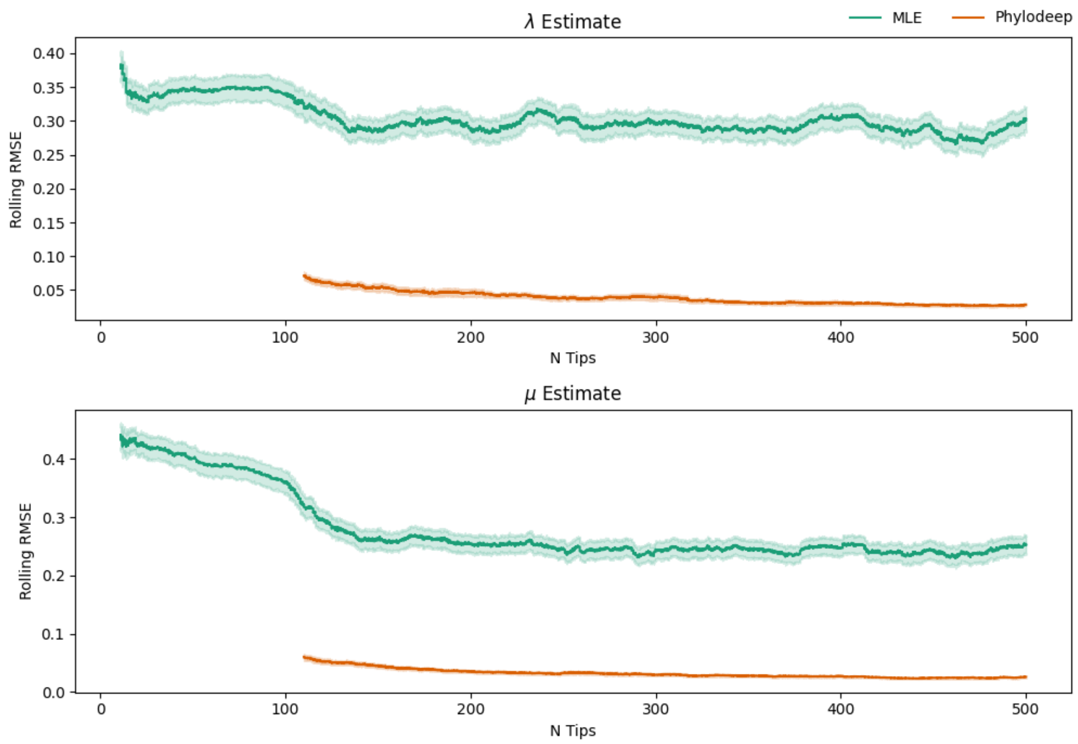

Figure 1: Bootstrap estimated RMSE of λ and μ estimates from MLE and Phylodeep models across varying tree sizes.

Urvi - talk here about your initial pass with a method of moments estimator and some difficulty, that's fine. Maybe give an overview of which packages you also tried for a closed form likelihood, and if you're comfortable and able, why those packages didn't quite work for us.

In the latest iteration, our MLE implementation is based squarely on the exact linear birth death-process likelihood from Phylopomp, that we have ported to Python as faithfully as possible, though this did not resolve our fitting issues. In fact, further reading of the 'treesimulator' documentation indicates that the proper likelihood actually comes from Stadler and Bonhoffer

5 Phylodeep

SANTOSH - model fitting process - talk a little bit here about the models from the Phylodeep paper, specifically the section **Parameter and model inference using neural networks** and the sections around it. Then, a quick one sentence blurb about why we chose to ignore the summary statistics neural network.

6 Results

Using the MLE and Phylodeep model together, we found that the Phylodeep model consistently outperformed the MLE method in estimating the main parameters (*mu* and *lambda*) of the Birth-Death trees. Surprisingly, the overall tree size does seem to reduce the RMSE for both Phylodeep and MLE methods, but not by much.

Our naive expectation was that the RMSE would be roughly comparable across these two methods. This, however, is not the case. Both methods do seem to improve with more data, but Phylodeep outperforms in terms of RMSE across all tree sizes. The Phylodeep methods were trained on median absolute percentage error, not RMSE, and so we cannot simply attribute its excellent performance to evaluation on an ideal target.

Figure **??** gives a clue as to what might be going on. We see a very sharp black line when we plot our fitted estimates of λ against μ. This is a clear indication of some issue in the MLE estimation, possibly independent of the actual likelihood calculation.

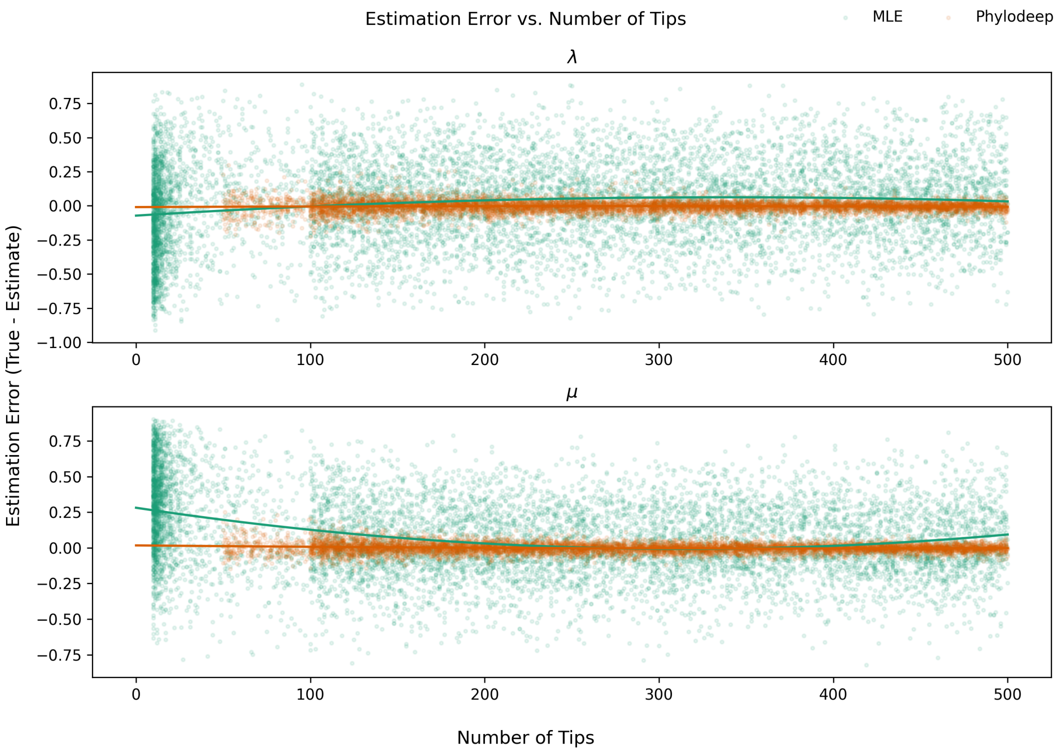

Figure 2: Raw error estimates with a quadratic polynomial fit overlaid for λ and μ from MLE and Phylodeep models. There appears to be no relationship with the size of the tree in this context, but the MLE is consistently biased upwards in small sample sizes for μ

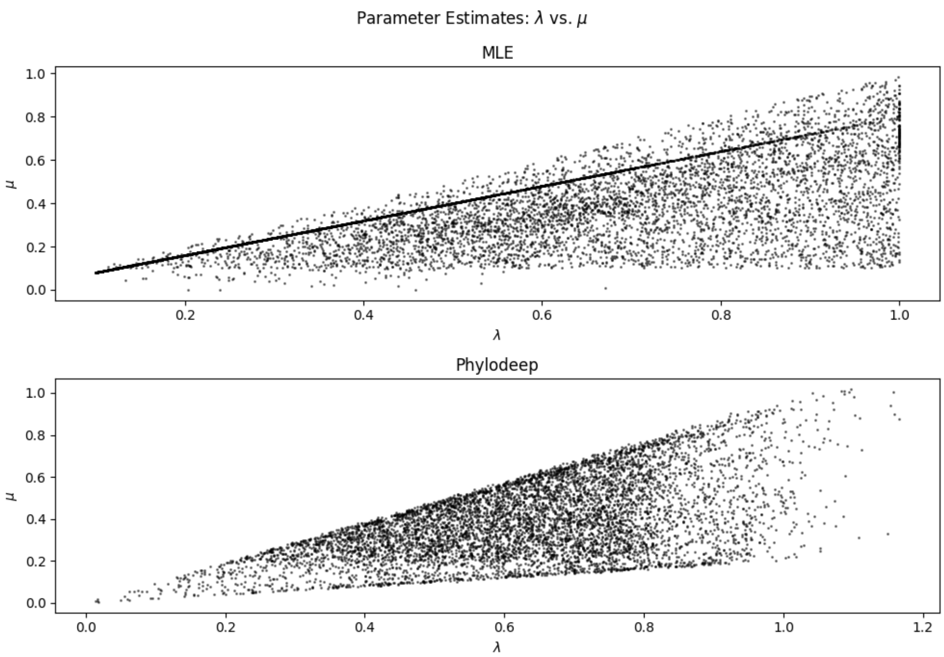

Figure 3: Scatter plot of λ vs μ estimates from MLE and Phylodeep models. The MLE estimates land on the lower boundary condition for μ close to 0, indicating an issue of identifiability with our current implementation.

When we consider the fitted values of the Phylodeep estimation vs. the MLE estimation, we see a pattern. Plotting the two parameter estimates together, we see that the MLE estimates often seem to settle on a boundary condition for mu, while the Phylodeep estimates do not have this issue. Further, Phylodeep estimates tend to miss equally both above and below the true parameter. The MLE has this issue in some parts of the curve (see Figure **??**), but we believe this indicates a problem with either our MLE implementation or the alignment of ‘treesimulator‘’s data dgeneration process and the likelihood function outlined by Stadler.

7 Conclusion

Our main research question was around the statistical efficiency of these two competing estimators with regards to data size (number of tips in the tree) in the context of the Birth-Death tree model. In our initial attempt at answering this question, it appears that Phylodeep outperforms the MLE in most contexts for all tree sizes.

But we encountered several issues that lead us to not be confident about our conclusions. First, our trees are generated exactly in the same manner as the Voznika paper, but it’s unclear that the likelihood we have identified correctly aligns with this model specification.

Solving these issues would allow us to more clearly identify the statistical tradeoffs between the MLE and deep learning approaches. In the future, we could expand our analysis to cover cases of slight misspecification in the deep learning models, for instance around the removal rate and sampling probability assumptions. Slight misalignments in the MLE likelihood seem like opportunities as well as issues, as being more precise about our data generating process would allow us to identify how robust the pre-trained deep learning models are in cases where the testing trees are not exactly aligned with the training data.